Fuzzy Logic

1. Fuzzy Logic system and neural network are combined to improve their individual performances ~~of the systems individual~~ over neuro-fuzzy system.

2. This has been ~~b~~ done to use the concept of human reasoning or human learning process.

3. This is possible due to the representation of initial knowledge by fuzzy rules based function and the degree of function approximation is iteratively by the learning capabilities of the neural network.

4. In neural-fuzzy system, the if-then conditions are represented by approximate weights and processing nodes.

5. The fuzzified neural network may include further fuzzy logic operators like ~~of~~ minimum, maximum, composite in addition to sum and product operators.

A classical set is a collection of objects of any kind, numbers, geometric points, chair, pencils etc.

Usually set is determined by naming all its members — The List method

OR, by some well defined properties satisfied by its members — The rule method

In the rule based method, the rule determines whether an element of universal set does or does not belong to the set under delimination.

$X \in A \Rightarrow$ an element/object 'X' is a member of set 'A' $\rightarrow$ X belongs to A

$X \notin A \Rightarrow$ 'X' is not an element of A $\rightarrow$ X does not belong to A

The List method ~~is~~ may not be applicable for the universal set, because this method is only applicable to finite set. However universal set mostly contains all the elements. ~~of particular class~~

Any set 'A' can be considered a ~~se~~ subset of 'U' $\rightarrow$ ~~$A \in$~~ $A \subset U$

$\phi \rightarrow$ is called 'null set' which contains nothing.

A ~~set~~ subset is considered from the ~~s~~ universal set. Now whether an element of universal set does or does not belong to that set, is the process of mapping universal set into the delimined set.

Individuals from universal set 'X' can be either member or non-member that can be defined by a function which is usually called ~~characteris~~ characteristics function.

For a given set 'A' this function assigns the value $\mu_A(x)$ to every $x \in X$

There can be two values of this function → TRUE or FALSE or in Boolean form

$$\mu_A(x) = \begin{cases} 1 & \text{if and only if } x \in \cancel{x} A \\ 0 & \text{if and only if } x \notin A \end{cases}$$

This function is also called Discrimination function. Delimining the discrimination function we divide the ~~undiv~~ universal set into two parts. The ~~delimination~~ function discrimination function is mapping of the area on which it ~~A~~ is delimined to the two-elements set.

Another method is membership function which is associated with element of the set 'A'.

$$\mu_A(x) = \begin{cases} 1 & \text{for } x \in A \\ 0 & \text{for } x \notin A \end{cases}$$

A classical set 'A', ~~witta~~ by a set of ordered pairs $(x, 1)$ or $(x, 0)$ which indicates $x \in A$ or $x \notin A$ respectively.

$$A = \{(x, 1) \mathrm{I} \ x \in X\}$$

$$A = \{(x, 0) \mathrm{I} \ x \notin X\}$$

The total number of elements in an universe (non fuzzy set) 'X' is called its cardinal number, denoted by 'n_x' where 'x' again is a individual level ~~for~~ in the universe.

Ideally an universal set should have cardinal number as infinite, because universal set is the collection of all possible elements.

Power Set

All possible sets of 'X' constitute a special set called power set $P(X)$

Say, $X = \{a, b, c\}$

$$P(X) = \{ \phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\} \}$$

The cardinal number ~~n_x~~ for universe X is $n_x = 3$

The cardinality of the power set, denoted by

$$n_{P(X)} = 2^{n_X} = 8$$

Fuzzy Set

Any set that empowers its members to have different grades of membership in an interval [0,1] is a fuzzy set.

The uncertainty was earlier handled by probability. The uncertainty means ~~who~~ the random process based characteristics and ~~as~~ also it indicates the process where occurrance of events strictly depends on the chance. This uncertainty does arise due to following reasons —

i) partial information about the problem

ii) information which is not fully reliable

iii) imprecision with the language through which the problem is defined

iv) more than one source of information are available.

Fuzzy set theory shows immense potential to solve such problems with uncertainty.

Fuzziness means vagueness.

Crisp & Fuzzy

A logic which demands a binary (0/1) type of handling is termed as crisp in the domain of fuzzy theory.

Ex- "Is water colourless?" → Yes/True (1) | No/False (0) — Ans

~~However~~ ~~Thus the situation~~ crisp means statements results in 0/1 situation

However, there are situation when/where the statements can not be confined to 0/1 statement only.

Ex- "Is Ram honest?" → Ans

Extremely honest : 1
honest at times : 0.4
very honest : 0.8
extremely dishonest : 0

In this situation, the answer can accept values in between 0 to 1 or either 0/1. Such a situation is termed as fuzzy.

A membership function $\mu_A(x)$ is associated with a fuzzy set $\bar{A}$ such that the function maps every element of the universe of disclosure X to the interval [0,1].

The mapping is represented $\mu_A(x) : X \to [0,1]$

If 'X' is universe of discourse and 'x' is a particular element of 'X', then fuzzy set 'A' defined on 'X' may be written as a collection of order pairs

$$A = \{(x, \mu_A(x)), x \in X\}$$

where each pair $(x, \mu_A(x))$ is called a ~~sign~~ Singleton.

Ex- Let, $X = \{g_1, g_2, g_3, g_4, g_5\}$ be the ref. set of students.

Let, $\bar{A}$ be the fuzzy set of smartness, where "smart" is a fuzzy linguistic term.

$$\bar{A} = \{(g_1, 0.4)(g_2, 0.5)(g_3, 0.9)(g_4, 1)(g_5, 0.8)\}$$

It ~~is~~ indicates smartness of g_1 is 0.4, smartness of g_2 is 0.5 and so on.

Membership function -

The membership function values need not not to be always discrete values. Quite ~~off~~ often these are described by continuous function.

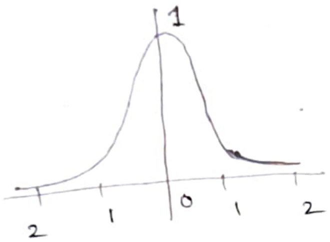
Membership function ~~is~~ can be represented by different ~~of~~ shape continuous functions -

Basically there are two methods to determine membership function (MF) —

i) use knowledge of human expert to specify MF in particular domain

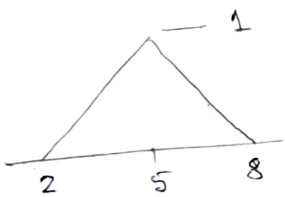
ii) defined from the collected database

a)



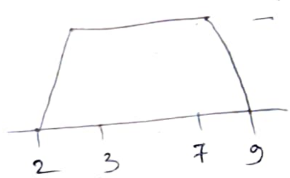
$$\mu_A(x) = \frac{1}{(1+x^2)}$$

b)



Triangular

~~$\mu_A(x)$~~



Trapizoidal

Ex - Consider the set of people in the following age group

0 - 10	40 - 50
10 - 20	50 - 60
20 - 30	60 - 70
30 - 40	70 and above

The fuzzy sets "young", "middle-age", "old" are represented by membership function graphs

Fuzzy Set operation

X – universe of discourse

$\bar{A}, \bar{B}$ – fuzzy sets

$\mu_A(x), \mu_B(x)$ – membership functions of ~~of~~ $\bar{A}$ & $\bar{B}$

Union

The union of two fuzzy sets $\bar{A}$ and $\bar{B}$ is a new fuzzy set $\bar{A} \cup \bar{B}$ also on X with membership function defined as

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$$
$$= \mu_{\underset{\sim}{A}}(x) \vee \mu_{\underset{\sim}{B}}(x)$$

Ex - In discrete form, x_1, x_2, x_3

$A = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0)\}$ ⇒ set of young

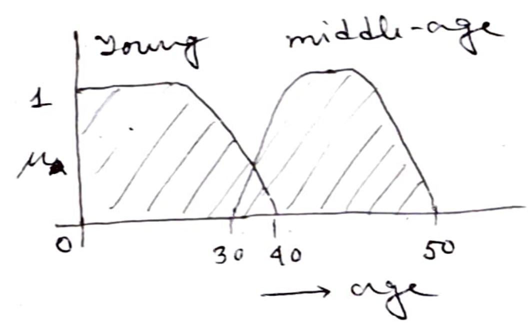
$B = \{(x_1, 0.8), (x_1, 0.2), (x_3, 1)\}$ ⇒ set of middle-age

$\bar{A} \cup \bar{B} = \{(x_1, 0.8), (x_2, 0.7), (x_3, 1)\}$

$\mu_{A \cup B}(x_1) = 0.8$

$\mu_{A \cup B}(x_2) = 0.7$

$\mu_{A \cup B}(x_3) = 1$



Intersection

The intersection of fuzzy sets $\bar{A}$ and $\bar{B}$ is a new fuzzy set $\bar{A} \cap \bar{B}$ with membership funⁿ is defined as

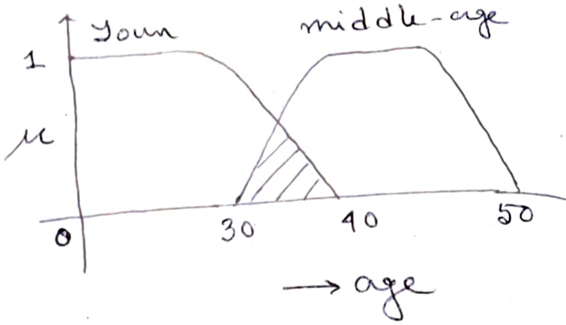
$$\mu_{\bar{A} \cap \bar{B}}(x) = \min(\mu_{\bar{A}}(x), \mu_B(x))$$
$$= \mu_{\underset{\sim}{A}}(x) \wedge \mu_{\underset{\sim}{B}}(x)$$

Ex – $\bar{A} \cap \bar{B} = \{(x_1, 0.5), (x_2, 0.2), (x_3, 0)\}$

$\mu_{\bar{A}\cap\bar{B}}(x_1) = 0.5$

$\mu_{\bar{A}\cap\bar{B}}(x_2) = 0.2$

$\mu_{\bar{A}\cap\bar{B}}(x_3) = 0$



Complement

The complement of a fuzzy set $\bar{A}$ is a new set $\bar{A}$ with membership function

$$\mu_{\bar{A}}(x) = 1 - \mu_A(x)$$

Ex – $\bar{A}$ is defined as "young"

The complement is "not young" → $\bar{A}^c$

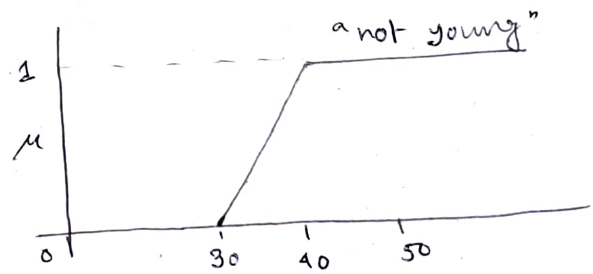
$\bar{A} = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0)\}$

$\bar{A}^c = \{(x_1, 0.5), (x_2, 0.3), (x_3, 1)\}$

$\mu_{\bar{A}^c}(x_1) = 0.5$

$\mu_{\bar{A}^c}(x_2) = 0.3$

$\mu_{\bar{A}^c}(x_3) = 1$



Products of two fuzzy set

The product of two fuzzy sets $\bar{A}$ and $\bar{B}$ is a new fuzzy set $\bar{A}.\bar{B}$ whose MF is

$$\mu_{\bar{A}.\bar{B}} = \mu_{\bar{A}}(x) \cdot \mu_{\bar{B}}(x)$$

Ex – $\bar{A} = \{(x_1, 0.2), (x_2, 0.8), (x_3, 0.4)\}$

$\bar{B} = \{(x_1, 0.4), (x_2, 0), (x_3, 0.1)\}$

$$\bar{A}.\bar{B} = \{(x_1, 0.08), (x_2, 0), (x_3, 0.04)\}$$

$\mu_{\bar{A}.\bar{B}}(x_1) = 0.08$

$\mu_{\bar{A}.\bar{B}}(x_2) = 0$

$\mu_{\bar{A}.\bar{B}}(x_3) = 0.04$

Equality

Two fuzzy sets are said to be equal $(\bar{A} = \bar{B})$ if ~~μA~~ $\mu_{\bar{A}}(x) = \mu_{\bar{B}}(x)$

Ex – $\bar{A} = \{(x_1, 0.2), (x_2, 0.8)\}$ $B = \{(x_1, 0.6), (x_2, 0.8)\}$

$\bar{C} = \{(x_1, 0.2), (x_2, 0.8)\}$

$$\bar{A} \neq \bar{B}$$

$$\mu_{\bar{A}}(x_1) \neq \cancel{\mu_{\bar{B}}(x_1)}\ \mu_{\bar{B}}(x_1)$$

But $\mu_{\bar{A}}(x_2) = \mu_{\bar{B}}(x_2)$

whereas, $\mu_A(x_1) = \mu_C(x_1)$

$\mu_A(x_2) = \mu_C(x_2)$

thus $\bar{A} = \bar{C}$

Product of fuzzy ~~s~~ set with crisp number –

Multiplying a fuzzy set by a crisp number a results a new fuzzy set product $a.\bar{A}$ with ~~f~~ MF

$\mu_{a.\bar{A}}(x) = a.\mu_A(x)$

$\bar{A} = \{(x_1, 0.4), (x_2, 0.6), (x_3, 0.8)\}$

$a = 0.3$

$a.\bar{A} = \{(x_1, 0.12), (x_2, 0.18), (x_3, 0.24)\}$

$\mu_{a.\bar{A}}(x_1) = 0.12$ $\quad \mu_{a.\bar{A}}(x_2) = 0.18$ $\quad \mu_{a.\bar{A}}(x_3) = 0.24$

Power of fuzzy

The α power of a fuzzy set $\bar{A}$ is a new fuzzy set A^α whose membership function is called Dilation

$\bar{A} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7)\}$

$\alpha = 2$

$\mu_{\bar{A}^2}(x) = (\mu_{\bar{A}}(x))^2$

$(\bar{A})^2 = \{(x_1, 0.16), (x_2, 0.04), (x_3, 0.49)\}$

$\mu_{\bar{A}^2}(x_1) = 0.16$ $\quad \mu_{\bar{A}^2}(x_2) = 0.04$ $\quad \mu_{\bar{A}^2}(x_3) = 0.49$

Difference

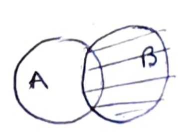
The difference of two fuzzy set $\bar{A}$ & $\bar{B}$ is $\bar{A} - \bar{B}$

$\bar{A} - \bar{B} = (\bar{A} \cap \bar{B}^c)$

$\bar{A} = \{(x_1, 0.2), (x_2, 0.5), (x_3, 0.6)\}$, $B = \{(x_1, 0.1), (x_2, 0.4), (x_3, 0.5)\}$

$\bar{B}^c = \{(x_1, 0.9), (x_2, 0.6), (x_3, 0.5)\}$

$\bar{A} - \bar{B} = (\bar{A} \cap \bar{B}^c)$



$= \{(x_1, 0.2), (x_2, 0.5), (x_3, 0.5)\}$

Disjunctive Sum

$$\bar{A} \oplus \bar{B} = (\bar{A}^c \cap B) \cup (\bar{A} \cap B^c)$$

$$\bar{A} = \{(x_1, 0.4), (x_2, 0.8), (x_3, 0.6)\}$$

$$B = \{(x_1, 0.2), (x_2, 0.6), (x_3, 0.9)\}$$

$$A^c = \{(x_1, 0.6), (x_2, 0.2), (x_3, 0.4)\}$$

$$B^c = \{(x_1, 0.8), (x_2, 0.4), (x_3, 0.1)\}$$

$$A^c \cap \bar{B} = \{(x_1, 0.2), (x_2, 0.2), (x_3, 0.4)\}$$

$$\bar{A} \cap B^c = \{(x_1, 0.4), (x_2, 0.4), (x_3, 0.1)\}$$

$$A \oplus B = \{(x_1, 0.4), (x_2, 0.4), (x_3, 0.4)\}$$

Crisp Relations

The concept of relations between sets (fuzzy) is built on the Cartesian product operator of sets.

The cartesian product of two sets A and B denoted by $A \times B$ is the set of all ordered pairs such that the first element in the pair belongs to A and the second element belongs to B.

$$A \times B = \left\{ \frac{(a, b)}{a \in A, b \in B} \right\}$$

Ex- $X = \{0, 1\}$ and $Y = \{a, b, c\}$

$$X \times Y = \{(0,a), (0,b), (0,c), (1,a), (1,b), (1,c)\}$$

$$X \times X = X^2 = \{(0,0), (0,1), (1,0), (1,1)\}$$

$$Y \times X = \{(a,0), (b,0), (c,0), (a,1), (b,1), (c,1)\}$$

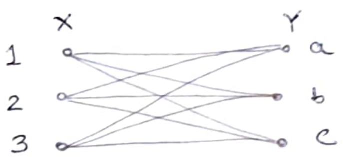
If $X \neq Y$ and X & Y are non-empty then $X \times Y \neq Y \times X$

Thus every element of universe X is related to every element of universe Y. The strength of the characteristics function is denoted by $\mathcal{X}$ where

- the value of unity is associated with a complete relationship
- the value of zero is related to no relationship.

$$\mathcal{X}_{X \times Y}(x, y) = \begin{cases} 1 & (x, y) \in X \times Y \\ 0 & (x, y) \notin X \times Y \end{cases}$$

when the universe or the sets are finite the relation can be represented by the relation matrix in the binary form.



$$R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \end{array}$$

The sagittal diagram ~~d~~ shows the points as the ~~ar~~ elements of universe and lines indicate relations between sets $X = \{1, 2, 3\}$ and $Y = \{a, b, c\}$

Crisp relation 'R' exists when matches between elements in two universes are constrained the characteristics function of relationship to map Cartesian space is

$X \times Y$ to binary values $(0, 1)$

$$\mathcal{X}_R(x, y) = \begin{cases} 1 & (x, y) \in R \\ 0 & (x, y) \notin R \end{cases}$$

Cardinality of crisp set

$$n_{X \times Y} = n_X \times n_Y$$

where n_X & n_Y are cardinality of X and Y

Given two relations R and S defined on $X \times Y$ and represented by relation matrices, the following operations can be done

Union : $R \cup S$

$$R \cup S(x,y) = \max(R(x,y), S(x,y))$$

Intersection : $R \cap S$

$$R \cap S(x,y) = \min(R(x,y), S(x,y))$$

Complement : $\bar{R}$

$$\bar{R}(x,y) = 1 - R(x,y)$$

Thus for binary relation $R(x,y)$ where $X = \{x_1, x_2, \ldots x_n\}$ and $Y = \{y_1, y_2, \ldots y_n\}$ the relation matrix R is two dimensional matrix where X represent rows and Y represents columns.

$$R(i,j) = 1 \text{ if } (x_i, y_j) \in R$$

$$R(i,j) = 0 \text{ if } (x_i, y_j) \notin R$$

Ex – Given $X = \{1, 2, 3, 4\}$

Let, the relation R be defined as

$$R = \{(x,y) / y = x+1,\ x, y \in X\}$$

$$R = \{(1,2), (2,3), (3,4)\}$$

Relation matrix
$$R = \begin{array}{c c} & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Composition of relations:

Given R to be a relation on X, Y and S to be a relation on Y, Z, then $R \circ S$ is a composition of relation on X, Z

$R \circ S = \{(x,z) / (x,z) \in X \times Z, \exists y \in Y$ such that $(x,y) \in R$ and $(y,z) \in S\}$

A common form of composition relation is the max-min composition

The relation matrices of the relation R and S the ~~min max~~ max-min composition is defined as

$$T = R \circ S$$

$$T(x,z) = \max_{y \in Y} (\min (R(x,y), S(x,z)))$$

Ex - let, R, S be defined on the sets $\{1,3,5\} \times \{1,3,5\}$

$R : \{(x,y) \mid y = x+2\}$, $S : \{(x,y) \mid x < y\}$

$R = \{(1,3), (3,5)\}$ $S = \{(1,3), (1,5), (3,5)\}$

The relation matrix are

$$R = \begin{array}{c} \\ 1 \\ 3 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 3 & 5 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{array} \qquad S = \begin{array}{c} \\ 1 \\ 3 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 3 & 5 \end{array} \\ \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

using max-min composition

$$T = R \circ S = \begin{array}{c} \\ 1 \\ 3 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 3 & 5 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

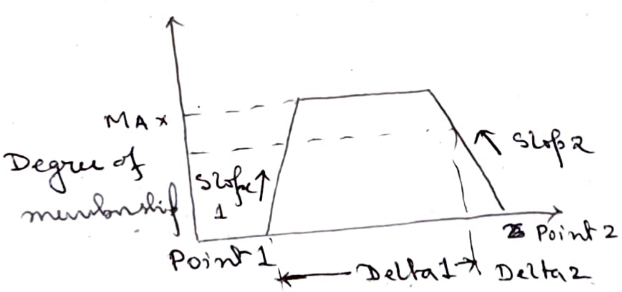
Take 1st row of R in coloum ~~and~~ alongwith 1st coloum of S.

Apply $T(x,y) = \max(\min(R(x,y), S(x,z)))$ to determine the 1st element of T.

R	S	$\min(R(x,y), S(x,z))$	$\max(\min(R(x,y), S(x,y)))$	
0	0	0	0	$\begin{bmatrix} 0 & & \\ & & \end{bmatrix}$
1	0	0		
0	0	0		
0	1	0	0	$\begin{bmatrix} 0 & 0 & \\ & & \end{bmatrix}$
1	0	0		
0	0	0		
0	1	0	1	$\begin{bmatrix} 0 & 0 & 1 \\ & & \end{bmatrix}$
1	1	1		
0	1	0		

~~Cardinality of fuzzy set~~

Ex Let consider an example of fuzzyfication of inputs to compute the membership ~~function~~

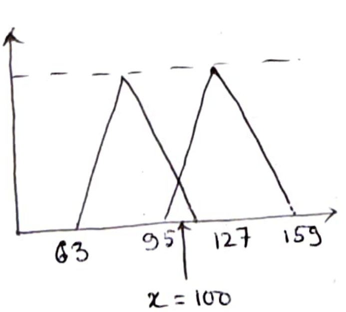


compute Delta 1 = x − point 1
Delta 2 = point 2 − x

If Delta 1 ≤ 0 or Delta 2 ≥ 0
Degree of membership 0

else

$$\min\begin{pmatrix} \text{Delta 1} * \text{Slope 1} \\ \text{Delta 2} * \text{Slope 2} \\ \text{Max} \end{pmatrix}$$



Delta 1 = 100 − 63 = 37

Delta 2 = 127 − 100 = 27

Slope 1 = $\frac{1}{32}$ = 0.03125

Slope 2 = $\frac{1}{32}$ = 0.03125

Degree of membership function = $\mu(x) = \min\begin{pmatrix} 37 \times 0.03125 \\ 27 \times 0.03125 \\ 1 \end{pmatrix}$

= 0.8438

To determine Fuzzy relations another method is Cosine Amplitude method.

Let, ~~ther~~ there are 'n' data samples and form a data array $X = \{x_1, x_2, \ldots, x_n\}$

Each of the data elements ~~samples~~, x_i, in the data array X is itself a vector of length 'm'

$$x_i = \{x_{i1}, x_{i2}, \ldots, x_{im}\}$$

Each elements of a relation, r_{ij}, results from a pairwise comparison of two data samples, x_i where strength of the relationship between data samples x_i and x_j is given by the membership value expressing the strength $r_{ij} = \mu_R(x_i, x_j)$

$$r_{ij} = \frac{\left|\sum_{k=1}^{m} x_{ik} x_{jk}\right|}{\sqrt{\left(\sum_{k=1}^{m} x_{ik}^2\right)\left(\sum_{k=1}^{m} x_{jk}^2\right)}} \quad \text{where } i, j = 1, 2, \ldots$$

Regions	x_1	x_2	x_3	x_4	x_5
x_{i1}	0.3	0.2	0.1	0.7	0.4
x_{i2}	0.6	0.4	0.6	0.2	0.6
x_{i3}	0.1	0.4	0.3	0.1	0

For example, $i = 1$ and $j = 2$

$$r_{12} = \frac{0.3 \times 0.2 + 0.6 \times 0.4 + 0.1 \times 0.4}{\left[(0.3^2 + 0.6^2 + 0.1^2)(0.2^2 + 0.4^2 + 0.4^2)\right]^{1/2}}$$

$$= \frac{0.34}{[0.46 \times 0.36]^{1/2}} = 0.836$$

$$R_1 = \begin{bmatrix} 1 & & & & \\ 0.836 & 1 & & \text{Sym} & \\ 0.914 & 0.934 & 1 & & \\ 0.682 & 0.6 & 0.441 & 1 & \\ 0.982 & 0.74 & 0.881 & 0.774 & 1 \end{bmatrix}$$

Max - min method

Another popular method like cosine amplitude method but simpler in calculation. It is different max-min composition method.

$$r_{ij} = \frac{\sum_{k=1}^{m} \min(x_{ik}, x_{jk})}{\sum_{k=1}^{m} \max(x_{ik}, x_{jk})} \quad \text{where } i,j = 1,2,\ldots n$$

$$r_{12} = \frac{\sum_{k=1}^{3} (\min(0.3, 0.2), \min(0.6, 0.4), \min(0.1, 0.4))}{\sum_{k=1}^{3} (\max(0.3, 0.2), \max(0.6, 0.4), \max(0.1, 0.4))}$$

$$= \frac{0.2 + 0.4 + 0.1}{0.3 + 0.6 + 0.4} = 0.538$$

$$R_1 = \begin{bmatrix} 1 & & & & \\ 0.538 & 1 & & & \\ 0.667 & 0.667 & 1 & & \\ 0.429 & 0.333 & 0.250 & 1 & \\ 0.818 & 0.429 & 0.538 & 0.429 & 1 \end{bmatrix}$$

Features of membership function

The core of a membership function for some fuzzy set $\tilde{A}$ is defined as the region of the universe that is characterised by complete and full membership in the set $\tilde{A}$.

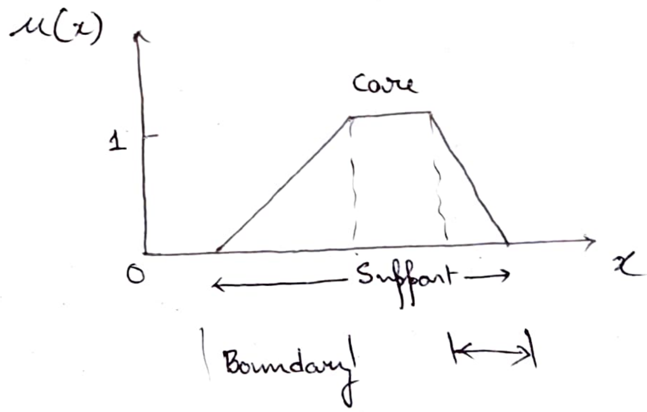
$\mu_A(x) = 1$

The <u>support</u> of a membership function for some fuzzy set $\bar{A}$ is defined as the region of the universe that is characterized by non-zero membership in the set $\bar{A}$.

$$\mu_A(x) > 0$$

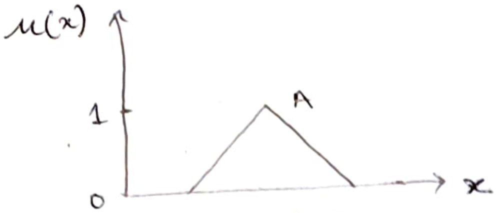
The <u>boundaries</u> of a membership function for some fuzzy set $\bar{A}$ are defined as that region of the universe containing elements that have nonzero membership but not complete membership.

$$0 < \mu_A(x) < 1$$

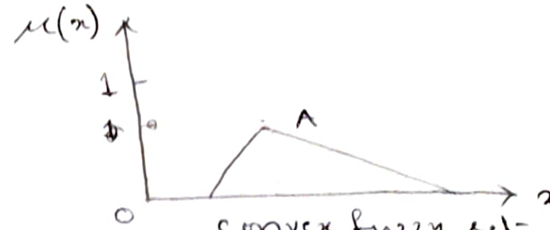


A <u>normal fuzzy</u> set is one whose membership function has atleast one element x in the universe whose membership value is unity.

For fuzzy sets where one and only one element has membership equal to one, this element is typically referred to as the <u>prototype</u> of the set.



Normal Fuzzy Set

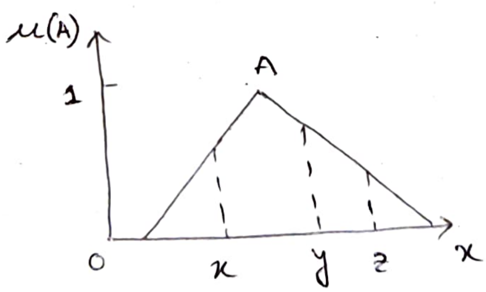


convex fuzzy set

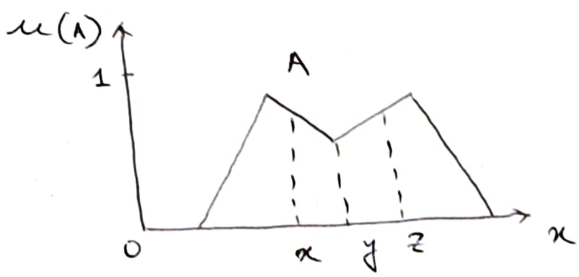
A convex fuzzy ~~is~~ set is described by a ~~mem~~ membership function whose membership values are strictly monotonically increasing or whose membership values are strictly monotonically decreasing.

The crossover points of a membership function are defined as the elements in the universe for which a particular fuzzy set A ~~has~~ has values equal to 0.5, i.e., $\mu_A(x) = 0.5$

The height of fuzzy set A is maximum value of the membership function, i.e., $\max\{\mu_A(x)\}$ If height of fuzzy is less than unity, then the fuzzy set is said to be subnormal.



Convex, Normal Fuzzy Set

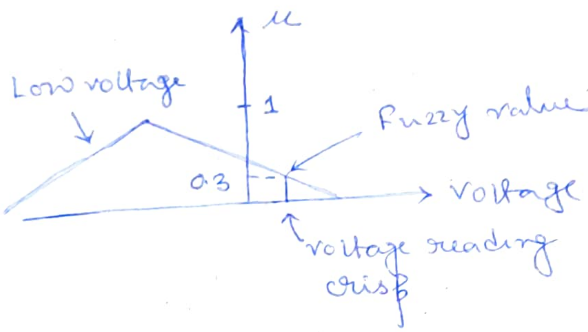


Nonconvex, normal fuzzy set

Fuzzification

Fuzzification is the process of making a crisp quantity fuzzy. We do this by simply recognizing that many of the quantities that we consider to be crisp and deterministic are actually non-deterministic ~~at~~ at all: They carry considerable uncertainty.

Let, we consider an example of fuzzyfication ~~for~~ from crisp data.



In this figure we might want to compare a crisp voltage reading to a fuzzy set "Low voltage".

In the fig. the crisp voltage intersects the fuzzy set "Low voltage" at a membership of 0.3., i.e, fuzzy set and the reading can be said to agree at a ~~m~~ membership value of 0.3.

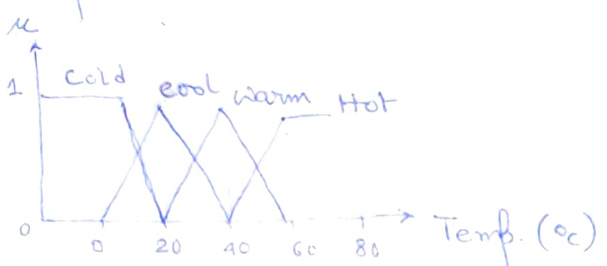
There are some methods for ~~asso~~ assigning membership values ~~ad~~ or functions to fuzzy variables -

(i) Intuition

This is derived based on human intelligence. Intuition involves contextual and semantic knowledge about an issue.

Let us take an example,

consider the membership function for fuzzy variable temperature.



The figure shows various shapes on the universe of temperature as measured in units of °c.

Here each curve is ~~is~~ a membership function corresponding to fuzzy variables — cold, cool, warm and hot. Now, these temperature curves are ~~ruff~~ referred to the range of human comfort and if they are referred to the range of safe operating temperatures for steam turbine, then the curves will change.

The important characteristics of these curves are (in fuzzy operation) the fact that they overlap.

The precise shape are not important for the curves, rather the approximate placement is important.

Inference

In this ~~mtn~~ method, knowledge is used for deductive reasoning, i.e., to infer a conclusion from a given body of facts and knowledge. There are many methods, but the formal knowledge on geometry and geometric shape ~~are~~ is illustrated.

In the identification of a triangle, let

A, B, C be the inner angles of the triangle, such that $A \geq B \geq C$ and $A + B + C = 180°$

$$U = \{(A, B, C) \mid A \geq B \geq C \geq 0;\ A + B + C = 180°\}$$

A number of geometric shapes are defined that can identify collection of angles for the given equation

$\underset{\sim}{I}$ Approx. isosceles triangle
$\underset{\sim}{R}$ Approx. right angle triangle
$\underset{\sim}{IR}$ Approx. isosceles and right ~~angle~~ triangle
$\underset{\sim}{E}$ Approx. equilateral triangle
I other triangles.

The membership values for all these triangle are inferred as the knowledge of geometry helps to make the membership assignments.

* Approx. isoscele triangles membership -

$$\mu_{\underset{\sim}{I}}(A, B, C) = 1 - \frac{1}{60°} \min(A - B,\ B - C)$$

* Fuzzy right triangle membership -

$$\mu_{\underset{\sim}{R}}(A, B, C) = 1 - \frac{1}{90°} |A - 90°|$$

* Approx. isosceles and right triangle membership -

$$\underset{\sim}{IR} = \underset{\sim}{I} \cap \underset{\sim}{R}$$

$$\mu_{\underset{\sim}{IR}}(A, B, C) = \min[\mu_{\underset{\sim}{I}}(A, B, C),\ \mu_{\underset{\sim}{R}}(A, B, C)]$$

$$= 1 - \max\left[\frac{1}{60°} \min(A - B,\ B - C),\ \frac{1}{90°} |A - 90°|\right]$$

* Fuzzy equilateral triangle membership –

$$\mu_{\underset{\sim}{E}}(A, B, C) = 1 - \frac{1}{180^\circ}(A - C)$$

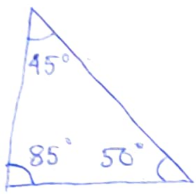
* Other triangle membership –

$$\underset{\sim}{I} = = (\underset{\sim}{I} \cup \underset{\sim}{R} \cup \underset{\sim}{E}) = \underset{\sim}{I} \cap \underset{\sim}{R} \cap \underset{\sim}{E}$$

$$\mu_{I}(A, B, C) = \min\{1 - \mu_{I}(A, B, C),\ 1 - \mu_{E}(A, B, C);\ 1 - \mu_{R}(A, B, C$$

$$= \frac{1}{180^\circ} \min\{3(A - B),\ 3(B - C),\ 2|A - 90^\circ|,\ A - C\}$$

Ex –



Define a specific triangle

$$\{X : A = 85^\circ \geq B = 50^\circ \geq C = 45^\circ, \text{ where } A + B + C = 180^\circ\}$$

$$\cancel{\mu_R}\ \ \mu_{\underset{\sim}{I}}(A, B, C) = 1 - \frac{1}{60^\circ}\min(A - B,\ B - C)$$

$$= 1 - \frac{1}{60^\circ}\min(85^\circ - 50^\circ,\ 50^\circ - 45^\circ)$$

$$= 1 - \frac{1}{60^\circ} \times 5$$

$$= \frac{11}{12} = 0.916$$

$$\mu_{\underset{\sim}{R}}(A, B, C) = 1 - \frac{1}{90^\circ}|A - 90^\circ|$$

$$= 1 - \frac{5}{90^\circ}$$

$$= 1 - \frac{1}{18} = 0.944$$

$$\mu_{\underset{\sim}{IR}}(A, B, C) = \cancel{7}\ 0.916$$

$$\mu_{E}(A, B, C) = 1 - \frac{1}{180^\circ}(85^\circ - 45^\circ)$$

$$= 0.77$$

$$\mu_{T}(A, B, C) = \frac{1}{180^\circ}\min\{3 \times 35^\circ,\ 3 \times 5^\circ,\ 2 \times 5^\circ,\ 40^\circ\}$$

$$= 0.05$$

Thus the Triangle has ~~ha~~ the highest membership in ~~the~~ right triangle.

Rank ordering

A single individual, a committee, a poll, and other methods are used to assign membership values to a fuzzy variable. The preference is determined by pairwise comparisons and these determine the order of membership.

Ex — Say 1000 respond to a questionnaire about pairwise preferences among five colours

X = {red, orange, yellow, green, blue}

Define a fuzzy set ~~a~~ A as universe of colours "best colour"

	Red	orange	Yellow	Green	Blue	Total	%	Rank
Red	–	517	525	545	661	2,248	22.5	2
orange	483 ↙1000	–	841	477	576	2,377	23.8	1
Yellow	475	159	–	534	614	1,782	17.8	4
Green	455	524	466	–	643	2,087	20.9	3
Blue	339	524	386	357	–	1,506	15	5
						10,000		

From the above table we can find out that Red & orange are compaired in pair and 517 people preferred the colour red over orange. Such ~~a matrix~~ ~~is~~ ~~and~~ in the matrix the colour columns represent an "antisymmetric" matrix.

There are ~~to~~ total 10,000 responses for 10 comparisons.